

Deepfake Video Detection

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1. INTRODUCTION

Deepfakes are AI-generated videos that manipulate facial identity with high realism, raising concerns about misinformation and digital trust.

This work investigates a deep learning-based approach for detecting deepfake content using frame-level analysis, while also exploring **facial recognition** as a complementary signal.

2. DATASET & PREPROCESSING

- Dataset:** Celeb-DF [1] (celebrity deepfake dataset)
- Video-level split to prevent data leakage
- I-frames extracted from videos (variable number per video)
- Frames resized to 299x299
- Preprocessing using Xception normalization

No duplicate frames across train/validation/test splits (verified: 0 duplicates across 209,278 images).

3. METHODOLOGY (DEEP LEARNING)

Model: Xception (pretrained on ImageNet) [2]

Two-stage training:

- Feature extraction (frozen base)
- Fine-tuning (unfrozen layers)

Input: individual video frames (I-frames)

Task: Binary classification (Real vs. Fake) at the frame level

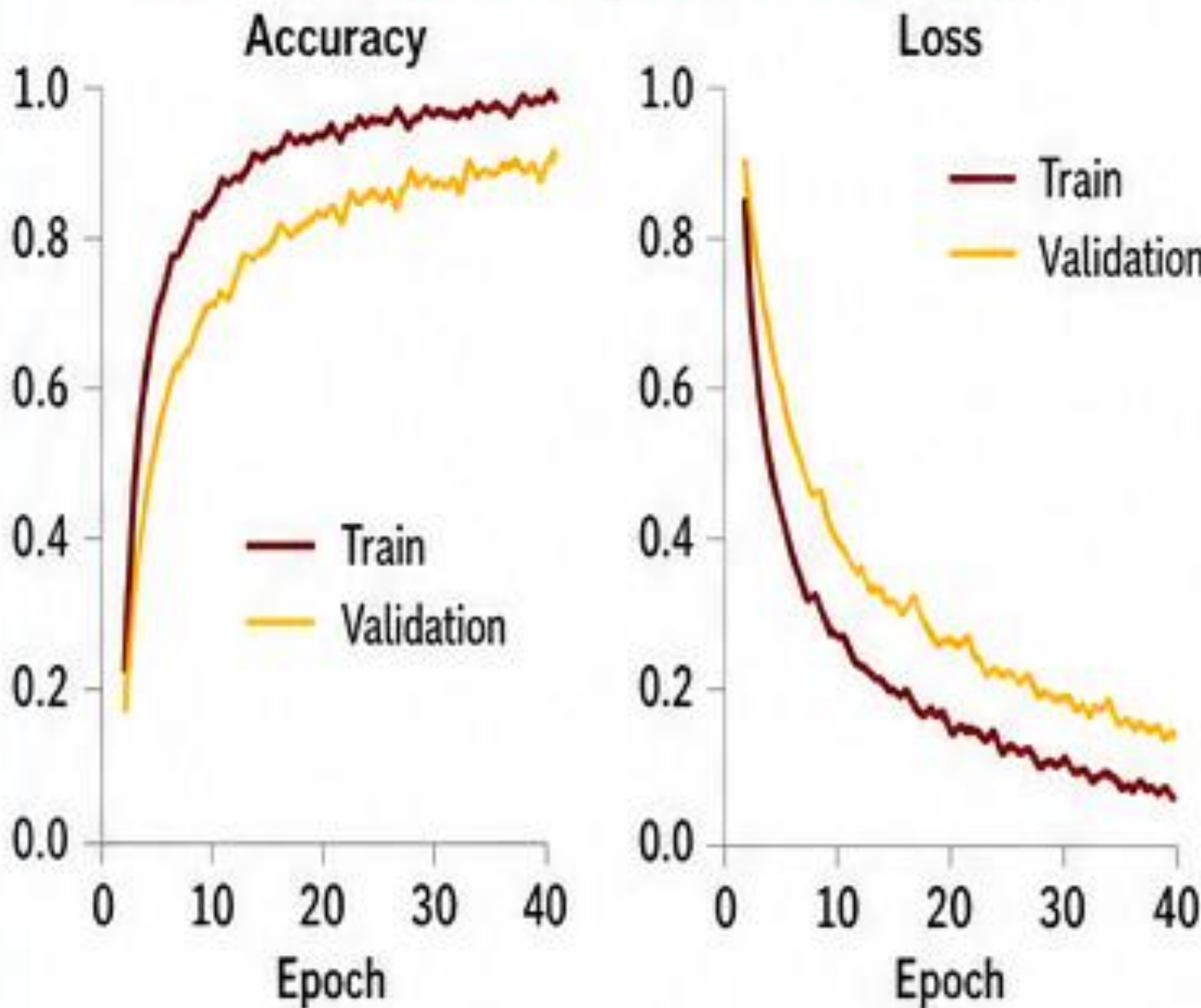
REFERENCES

- [1] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for DeepFake forensics," arXiv:1909.12962, 2019.
- [2] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," Proc. CVPR, 2017.

4. RESULTS (TEST SET PERFORMANCE)

Accuracy	Balanced Accuracy	AUC (ROC)	Fake Recall (Sensitivity)	Real Recall (Specificity)
94%	88.24%	0.97	97%	80%

TRAINING CURVES (40 EPOCHS TOTAL)



CONFUSION MATRIX (TEST SET)

		Predicted	
		Fake	Real
Actual	Fake	26,028	835
	Real	953	3,717

Key Insight:
More false positives (real → fake), indicating bias toward detecting manipulation.

CLASSIFICATION REPORT (TEST SET)

Class	Precision	Recall	F1-Score	Support
Fake	0.96	0.97	0.97	26,883
Real	0.82	0.80	0.81	4,670
Accuracy				94%
Balanced Accuracy				88.24%
AUC (ROC)				0.97
Macro Avg (F1)				0.89
Weighted Avg (F1)				0.94

5. QUALITATIVE RESULTS (EXAMPLE PREDICTIONS)

1. Real → Real (Correct)



GT: Real Pred: Real
Conf: 0.99

2. Fake → Fake (Correct)



GT: Fake Pred: Fake
Conf: 0.98

3. Real → Fake (Error)



GT: Real Pred: Fake
Conf: 0.72

4. Fake → Real (Error)



GT: Fake Pred: Real
Conf: 0.61

Examples of correct and incorrect predictions. Misclassifications typically occur in challenging visual conditions or subtle manipulations.

6. LIMITATIONS

- Frame-based approach ignores temporal consistency.
- Class imbalance biases predictions toward "fake".
- Possible identity overlap across dataset splits.
- Performance may degrade under heavy compression or severe image quality degradation.

7. FUTURE WORK



Temporal Modeling
(Video-level)

Incorporate temporal information using 3D CNNs, Transformers, or RNNs for stronger video-level detection.



Robustness & Generalization

Evaluate under real-world distortions (compression, noise, blur) and cross-dataset generalization.



Facial Recognition
(Exploratory Extension)

Leverage identity consistency checks across frames as a complementary signal for detection.